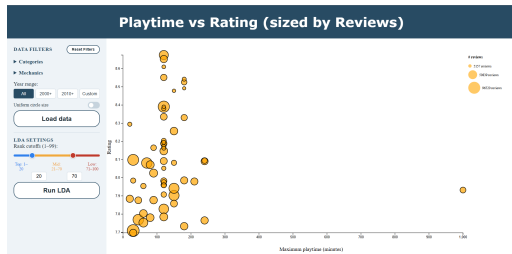


Visual Analytics: Project 1

Board Game Analysis

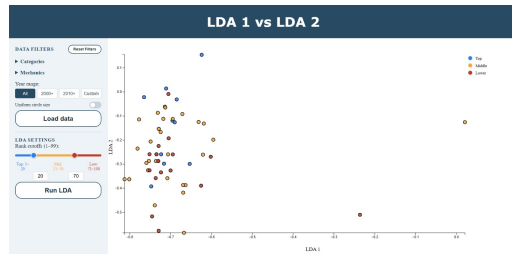
Task 2.1: Explore how maximum playtime relates to ratings and review counts across board games

Task 2.2: Compare rank-based groups using LDA dimensionality reduction



Yeganeh: Brainstorming (5 h), Scatterplot, LDA, year filter, rank slider, most visual design (40 h)

Görkem: Brainstorming (5 h), Category & Mechanics filter



Gaurav: Brainstorming (5 h), Auto-refresh LDA and zoom-in (5 h), Editing presentation (5 h)

Hassan: Brainstorming (5 h)

Why these tasks?

- **Task 2.1 (Explore / Correlate):**
“Does maximum playtime predict rating and commercial reach?”
⇨ A developer planning game length needs to know if longer games score higher and reach more players.
- **Task 2.2 (Compare / Group):**
“Do top-ranked games have systematically different design characteristics?”
⇨ If top-ranked games, cluster separately, there is a learnable design profile useful for a game developer.

Task 2.1 - 5-tuple

Goal : *Explore*

Means : *Identify*

Characteristics : *Correlate*

Target : *maxplaytime , rating ,
num_of_reviews*

Cardinality : *Overall*

Task 2.2 - 5-tuple

Goal : *Explore*

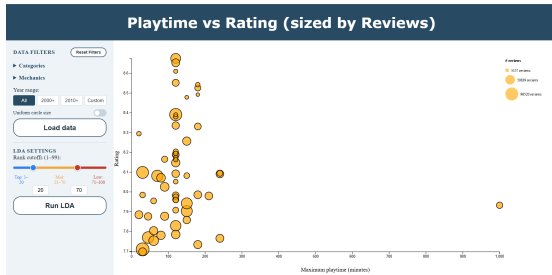
Means : *Compare*

Characteristics : *Group*

Target : *year , minage , minplayers,
maxplayers, minplaytime, maxplaytime*

Cardinality : *Overall*

Task 2.1: Playtime vs. Rating



Visualization: Scatterplot

x-axis Maximum playtime (min)
y-axis Rating (d3.extent)
circle r # reviews (scaleSqrt)
colour Orange (uniform)

Preprocessing:

Nested JSON → flat object:
{title, maxplaytime, rating,
num_of_reviews}

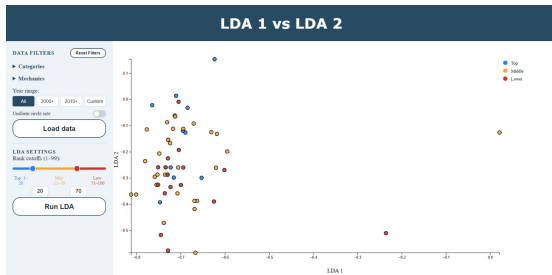
Included: size legend, tooltip

Excluded: colour groups, grid lines

Task 2.1: Why this visualization supports the task

- **Scatterplot for Correlate:** Position is the most accurate visual representation for two quantitative attributes. A visible trend or spread in the point cloud directly answers the *Correlate* characteristic.
- **scaleSqrt for circle size:** We perceive area, not radius. A linear radius scale makes a 4x data value appear 16x visually. `scaleSqrt` makes area grow linearly, a more accurate encoding of review count.
- **d3.extent for y-axis:** Ratings span only 7.7–8.7. Starting at 0 compresses all variation into the top 10% of the chart.
- **No colour in Task 2.1:** There are no groups to distinguish. Colour here would imply conclusions that do not exist.
- **No grid lines:** The task is pattern detection, not precise reading. Grid lines add clutter without helping *Correlate*.

Task 2.2: Rank Groups



Visualization: LDA Scatterplot

x-axis LDA 1

y-axis LDA 2

colour Rank group

● Top (1–25) ● Mid (26–75) ● Lower (76–100)

LDA features (normalised to $[0,1]$):
year, minage, minplayers, maxplayers,
minplaytime, maxplaytime

Excluded: rating, num_of_reviews
Rank is derived from these, including them
is circular reasoning.

Task 2.2: Why this visualization supports the task

- **LDA for Compare + Group:** LDA is supervised, which means it finds the projection that *maximally separates* labelled groups. PCA would not guarantee group separation.
- **Feature normalisation:** year (≈ 2010) vs. minplayers (≈ 2) differ by 1000x. Without normalisation LDA is dominated by year. All features scaled to $[0, 1]$.
- **Colour legend included:** Without it, the three groups are unidentifiable.
- **Tooltip included:** There are overlapping circles; without it no individual game can be named.
- **Variable circle size excluded:** No meaningful 3rd attribute in this view. Variable size would add noise without encoding anything real.

Task 2.3: Interacting with LDA Parameters

Interactions implemented:

1. **Dual-handle rank cutoff slider**

Colour-coded track (blue/orange/red). Text inputs allow precise entry. Labels show exact ranges in real time. After entering the LDA view once, slider changes refresh the projection automatically.

2. **Year range preset buttons**

All / 2000+ / 2010+ / Custom.
98 of 100 games are from 2002–2021, so a continuous slider over 145 years (Crokinole, 1876) would be awkward.

3. **Category & Mechanics filters**

Checkbox lists with live search, applied server-side before preprocessing.

4. **Rectangular zoom-in**

Drag over a crowded region to zoom into that area. Double-click restores the full view.

Why low interaction cost?

- Slider drag **or** text field to set cutoffs
- One initial click: **Run LDA**
- Later slider changes update automatically
- Result updates in the same view with no need for reload

Why rank cutoff is the key parameter:

LDA is supervised, i.e, class labels are its most important input. Changing them redefines what “top” means. Top ≤ 10 asks a different question than Top ≤ 50 .

Demo

`http://localhost:3231`

1. **Load data** → explore scatterplot, hover over circles
2. **Run LDA** with default cutoffs (25 / 75)
3. **Adjust rank cutoffs** → see LDA change actively
4. **Filter by year** (2010+) → Load data → Run LDA

What could be improved?

- **Task 2.1:**

Brushing & linking - selecting games in the scatterplot highlights them in the LDA view, connecting both analyses.

- **Task 2.2:**

LDA feature importance - show which of the 6 features contribute most to LDA 1 and LDA 2, making the axes interpretable.

Game mechanics as LDA features, one-hot encoding of mechanics could better explain rank differences than design parameters alone.

- **General:**

Richer tooltip - add mechanics, categories, and player count so the viewer can explain why a game sits where it does.

Larger dataset - all 100 games are already top-100 BGG, so there is no “bad game” baseline for comparison.